

FRAN SÁNCHEZ RODRIGO

[Unreal Engine] Tools, Engine & Optimization Engineer

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EXPERIENCE

Unreal Tools & Engine Programmer Unreal Console Porting Specialist at Catness Game Studios

Dec 2021 - Mar 2025

- Game porting with Unreal Engine 4 and 5 to console platforms:
 - [Xuan Yuan Sword 7](#) (Switch)
 - [Metamorphosis VR](#) (Oculus Quest 2)
 - [American Arcadia](#) (Switch, PS4)
- Plugin and tool development to expand the engine base functionality:
 - Easily applicable custom device profiles at runtime.
 - Project analysis tools and statistics display.
 - Custom blueprint nodes.
 - Context menu extensions with custom tasks.
- Plugin and tool development to ease the porting process to consoles:
 - Xbox user management controller handling.
 - PS5 Activities management.
 - Optimization tools (automatic MaxDrawDistance, tick management, mesh instancing).
 - Automatic asset replacement by platform.
- General engine modifications and fixes:
 - Indexed Actor tags optimization.
 - Button auto-hover for consoles.
 - Mobile Software Occlusion fixes.
 - Improved behavior of MinLOD for consoles.
 - Build deployment with custom folder structure and visible build hashes.

Quality Assurance Technician at Lollipop Robot

Feb 2016 - Sep 2020

- Playtesting and Compliance testing for PC and consoles (Xbox One, PS4, Switch).
- Experience using popular bug trackers like Jira, Mantis, Bitbucket, or GitHub.

EDUCATION

Master's Degree in Video Game Programming

2020 - 2021

U-Tad (Madrid)

Graduate (Bachelor's Degree) in Video game Design and Development

2012 - 2016

Universitat Jaume I (Castellón)

SKILLS AND COMPETENCIES

- Advanced level of game programming in Unreal Engine C++.
- Intermediate level of game development in Unity.
- High-level experience using version control systems (Git, Perforce).
- Fast adaptability to new codebases.
- Ability to work with low level SDKs and systems.
- Experience with different programming and scripting languages: Java, C#, Python, LUA.
- Basic knowledge of CMake for C++ development.

LANGUAGES

- English: Professional working proficiency.
- Spanish: Native or bilingual proficiency.
- Catalan: Native or bilingual proficiency.

ACADEMIC PROJECTS

Ronin-Hood Sisters

Master's Degree in Video Game Programming

Tactical RPG made with Unreal Engine 4 (C++), based on Magical Girls in costumes with elemental powers and destructive abilities.

Worked programming the game world, a tool to create the maps, fog of war, save system, and the logic behind multiple UI, like the costume selection screen, the inventory and the combat GUI.

- [Detailed explanation of all my work in the game](#)

Medieval Floor

Master's Degree in Video Game Programming

Action game made with Unreal Engine 4 (C++). Cooperate locally in a group of 4 players to clear waves of increasing difficulty.

Worked programming the gameplay flow (generating enemy pools and spawning them in waves), and saving the high score to disk.

- [Detailed explanation of all my work in the game](#)
- [Check the source code](#)

Unity Zombie Killer

Master's Degree in Video Game Programming

First-person zombie survival game where you have to complete 3 different quests to win. Made with Unity (C#) in 5 days.

Worked programming the weapons, pickable items, zombies' AI, and the quest system.

- [Detailed explanation of all my work in the game](#)
- [Check the source code](#)

OTHER PROJECTS

Unused Node Cleaner

"Unused Node Cleaner" is an Unreal Engine plugin that allows you to clean your blueprints from unused nodes that someone left somewhere and are still there to distract you.

- [Fab.com download page](#)

Quantum fields

Entry for the 2023 Epic MegaJam. Portal-like game where you open portals to see the past.

Implemented an async dynamic mesh cut system.

- [Detailed explanation of all my work in the game](#)

Urkaz Moon Tools

A Minecraft Java Edition mod that adds two items to check the moon phase. Its main purpose is to be used in conjunction with other mods. Written in Java.

- [Github repository](#)